

HINTS AND SOLUTIONS

Q.1 A

Q.2 B

Sol. $W = \frac{1}{2} \times 3v_0 \times P_0 + 3v_0 \times P_0$

$$W = \frac{3}{2} P_0 v_0 + 3v_0 P_0$$

$$W = \frac{9}{2} P_0 v_0$$

$$\Delta U = nC_v \Delta T = n \left(\frac{3R}{2} \right) (T_f - T_i)$$

$$\Delta U = \frac{3}{2} nR (T_f - T_i) = \frac{3}{2} [P_f V_f - P_i V_i]$$

$$\Delta U = \frac{3}{2} [4P_0 V_0 - 2P_0 V_0]$$

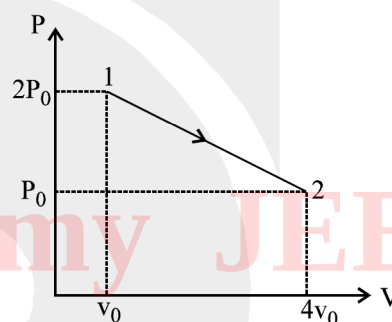
$$\Delta U = 3P_0 V_0$$

$$\Delta Q = \Delta U + W = \frac{9}{2} P_0 V_0 + 3P_0 V_0 = \frac{15}{2} P_0 V_0$$

$$P_0 = \frac{10^6}{2}, V_0 = 0.1$$

$$\Delta Q = \frac{15}{2} \times \frac{10^6}{2} \times 0.1 = 375000 \text{ J}$$

$$\Delta Q = 375 \text{ kJ}$$



Q.3 B

Sol. $n = \frac{PV}{RT}$

Final pressure & temperature is same hence volume will be in ratio of number of moles.

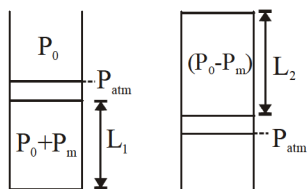
Q.4 D

Sol. $(76 - 67) \rho g \times (19 \times 1) = 76 \rho g V$

$$V = \frac{9}{4} = 2.25$$

Q.5 B

Sol. $P_m = \text{Pressure due to mercury.}$



For constant, 'T'

$$(P_0 + P_m) L_1 A = (P_0 - P_m) L_2 A \quad \dots (i)$$

$$P_m = \frac{mg}{A} = \frac{0.0029}{20 \times 10^{-6}} = 10^3 \text{ Pa}$$

So, from (i), $\frac{L_2}{L_1} = \frac{101}{99}$

Q.6 A

Sol. $\Delta \text{K.E.} = \Delta U$

$$\frac{1}{2} \times n \times 30 \times 10^{-3} \times (100)^2 = n \times \frac{5}{2} R \times \Delta T$$

$$\Delta T = \frac{60}{R}$$

Q.7 B

Q.8 C

Q.9 A

Sol. $Vp^2 = C$

$$\beta = -v \left(\frac{dp}{dv} \right)$$

$$p^2 + 2vp \frac{dp}{dv} = 0$$

$$\frac{dp}{dv} = \frac{-p}{2v}$$

$$\beta = -v \times \left(\frac{-p}{2v} \right) = \frac{p}{2}$$

Q.10 D

Q.11 B

Sol. $C_1 = \frac{R}{\gamma-1} - \frac{R}{m-1} = \frac{R}{\gamma-1} + \frac{R}{2} = C_v + \frac{R}{2} = \frac{C_p + C_v}{2}$

$$C_2 = \frac{R}{\gamma-1} - \frac{R}{0.5-1} = \frac{R}{\gamma-1} + 2R = C_v + 2R = C_p + R$$

$$C_3 = -\infty$$

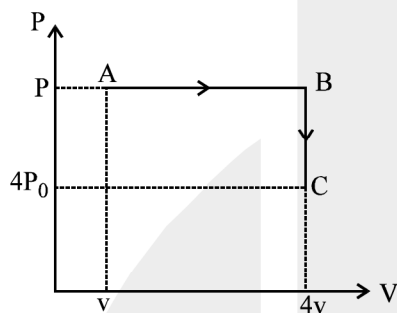
$$C_4 = \frac{R}{\gamma-1} - \frac{R}{1.25-1} = C_v - 4R = -ve$$

$$C_5 = \frac{R}{\gamma-1} - \frac{R}{\gamma-1} = 0$$

- Q.12 B
 Q.13 ACD
 Q.14 ACBD
 Q.15 BD
 Q.16 BC
 Sol. as $f \uparrow$, $C \uparrow$

assuming $P \uparrow$ as $T \uparrow$, $\frac{1}{P} \downarrow$, compressibility $\downarrow$.]

- Q.17 AC
 Q.18 ABC
 Sol. (A), (B) (temp $\downarrow$) (C) $Q_1 > Q_2$



- Q.19 [Ans. (A)-PQ, (B)-RS, (C)-PQ, (D)-RS]

- Sol. (A) Isothermal process, $\Delta V = 0$, $W = +ve \Rightarrow \Delta Q = +ve$
 Since volume is increasing in both processes
 (B) A is adiabatic with $-ve$ work done $\Rightarrow \Delta U = +ve$
 B is isothermal
 (C) Same as A
 (D) Same as B

- Q.20 [Ans. 0004]

Sol.
$$\left. \begin{aligned} P_0 &= \frac{k\ell}{A} \\ 2P_0 &= \frac{k\ell'}{A} \end{aligned} \right\} \Rightarrow \ell' = 2\ell$$

$$T' = \frac{2P_0(2\ell A)}{R} = \frac{2P_0 2V_0}{R} = \frac{4P_0 V_0}{R}$$

$$= 4T_0 = nT_0$$

- Q.21 [Ans: $3.2 \times 10^4 \text{ N/m}^2$, $150\sqrt{2} \text{ K}$, $1.52 \times 10^4 \text{ J}$, $-1.52 \times 10^4 \text{ J}$]